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ORIGINAL ARTICLE

The curvature modeling for the double layered clad plate by asymmetric rolling with different diameters

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Abstract

The plate curvature model plays an important role in the prediction and control of the double layered clad plate. However, there is currently no such a fast, precise and stable plate curvature model, and the actual shear stress on the cross section is uneven distributed actually which is conflicted with the uniform assumption of the traditional model. Based on the non-uniform distribution, this paper innovatively presents a theoretical plate curvature model using the flow function method and the upper bound method together without the complex stress analysis. The inlet boundary conditions of the model are more accurately modified and optimized as nonlinear functions. The velocity field and strain rate field were built depending on the flow function field. The plastic deformation, shear and friction power were developed by the five-node Gauss–Legendre quadrature method. The post-rolling strain model was obtained by integrating the positive strain rate and shear strain rate with time. The curvature model was constructed by calculating the curvature due to linear and shear strain differences. To demonstrate the validity of the theoretical model, the experiments and the simulation were conducted. The results showed that the relative deviation of the numerical values v_a and v_b is less than 10.0%, accounting for 95.1% and 95.4%, compared with theoretical values. The shear power difference between layer a and b is the main reason for bending. The bending phenomenon appeared at the outlet section of the plate, and the velocity difference provided the shear stress required for bending. The increase of the diameter ratios leads to the increase of the shear strain difference and shear power, and it leads to an increase in the total curvature. To certify the precision of the curvature model, the deviation of the theoretical peak curvature values is 12.05%, and the average deviation is 5.92%, compared with the simulation and experiments. The model can quickly grasp the affecting law of key rolling process parameters on curvature, which will provide theoretical and technical basis for rolling process.

Keywords Plate curvature · Upper bound theorem · Clad plate · Flow function · Strain

Symbol table

R_1, R_2, R_j	Radius of the top and bottom rolls and the bonding surface
t_i, t_{ia}, t_{ib}	Total, cladding and base layer thickness before rolling
t_{xa}, t_{xb}	The thickness at any position in the deformation region
t_f, t_{fa}, t_{fb}	Total, cladding and base layer thickness after rolling
Δh_1, Δh_2, Δh_j	The reduction of top and bottom rolls and the deformation bonding surface
l_1, l_2, l_j	Length of top and bottom rolls and bonding surface in the deformation region
l_a	The length of the upper surface of the rolled part after being elongated by bending deformation
l_b	The length of the lower surface of the rolled part after being compressed by bending
l_0	The original length of the rolled part
ω_1, ω_2	Speed of top and bottom rolls
$\phi 0\ a$, $\phi 0\ b$	Basic flow functions
Q_a, Q_b	Volume flow per unit area
$\phi^*\ a$, $\phi^*\ b$	Added flow functions
ϕ_a, ϕ_b	Optimal flow functions

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Γ_{sa}, Γ_{sb}	Velocity discontinuities
y_1, y_2	Boundary functions along the top and bottom rolls
y_j	Contact boundary functions of cladding layer and base layer
y_3, y_4	The geometric equations of Γ_{sa} and Γ_{sb}
k_a, k_b	Shear yield strength
m	Friction factor
a_1, a_2, b_1, b_2	Streamline factor
α	The angle between the Γ_{sa} and horizontal direction
β	The angle between the Γ_{sb} and horizontal direction
via, v_{ib}	Inlet velocity
$\dot{\varepsilon}_{xa}$,$\dot{\varepsilon}_{xb}$	Horizontal strain rate components
ε_{xa}, ε_{xb}	The axial strain on the upper and lower surfaces of the clad plate
v_{xa}, v_{xb}	Horizontal velocity components
$\dot{\varepsilon}_{ya}$,$\dot{\varepsilon}_{yb}$	Vertical strain rate components
v_{ya}, v_{yb}	Vertical velocity components
$\dot{\varepsilon}_{xya}$,$\dot{\varepsilon}_{xyb}$	Shear strain rate components
W	Total power
W_{va}, W_{vb}	Plastic deformation power
$W_{\Gamma sa}$, $W_{\Gamma sb}$	Shear power
W_{fa}, W_{fb}	Frictional power
r_a, r_s	Plate radius due to shear strain and axial strain
r	Total radius

Subscript

a	Cladding layer
b	Base layer

1 Introduction

Magnesium alloy is currently one of the most light metals used in structural materials with a series of advantages, such as low density, high specific strength, damping, shock absorption and radiation resistance, which has a broad application prospect in the field of automobile and aerospace [1, 2]. However, the high price and the low corrosion resistance of magnesium alloy enormously restrict its usage and development. Meanwhile, aluminum alloy has gained attention due to its excellent properties, such as low density, high corrosion resistance, good electrical conductivity and thermal conductivity, etc. [3, 4]. The aluminum/magnesium clad plate formed by attaching the aluminum alloy plate with excellent corrosion resistance to the surface of magnesium alloy combines the excellent properties of each of the two metals [5]. This layered metal has been widely used in important fields such as aerospace, mechanical and electrical power, transportation, marine ships and artificial satellites. In order to make the deformation of the double layer metal more coordinated, asymmetric rolling is proposed as an asymmetric rolling process. Compared with symmetric rolling, asymmetric rolling provides a shear stress in the rolled area by adjusting the diameter or speed of the top and bottom rolls. It can change the stress and strain distribution of the plate, accelerate the extrusion of fresh metal, while the interface generates a lot of heat to accelerate the movement of metal atoms at the interface and refine the grain [6]. In recent years, asymmetric rolling is mainly used to study the microstructure and forming properties of metal composites [7, 8].

During asymmetric rolling, however, the bending problem can be induced by various asymmetric factors, and the resulting bending behavior may be more complex than the conventional rolling. In order to gain the adequate flat plates, controlling the rolling parameters accurately and improving the synergistic deformability between the diffusion elements at the bonding interface are indispensable. Liang et al. [9] imitated the asymmetric plate rolling with different diameter ratios by simulation method and analyzed the validity of roller diameter as a method for controlling plate curvature through experiments. Salganik et al. [10] investigated the deformation mechanism of the different roll speeds and integrated the influence of three asymmetric factors on the plate shape. Qwamizadeh et al. [11] investigated the plate curvature of double-layer plates after asymmetric rolling. Based on finite element method, Yang et al. [12] summarized the influences of the work roll offset, reduction and speed ratio on the plate curvature. The numerical results of the above study can provide a wide range of rolling parameters for the curvature prediction and control during the asymmetric rolling. However, in order to improve the computational speed and calculating precision, the theoretical model of plate curvature needs to be built by using an appropriate mathematical solution method.

The rolling composite process of double-layered metal plates involves material asymmetry and geometric asymmetry [13], which may pose many difficulties for the in-depth study of the mechanism. For controlling rolling parameters and predicting curvature, many scholars have developed some analytical models by the slab method, such as curvature and strain. In order to compute the plate curvature, Jiang et al. [14, 15] set up a theoretical model using the slab method. Afrouz et al. [16] researched the mechanical parameters and curvature for asymmetric rolling of the cold rolled strip with uneven shear stresses using the slab method. However, in order to gain more detailed information parameters faster, the flow function method and the upper bound method also provide effective and feasible ideas for the prediction of plate curvature. The velocity and stress–strain field can be calculated quickly without complex stress analysis.

Ding et al. [17] gained the velocity field of the fixed angle of the chamfered edge using the flow function method, derived the dog-bone shape coefficient from the volume invariant condition and then obtained the plastic deformation power, shear force and friction force separately using the equal yield criterion. Jiang et al. [18] modified the boundary conditions of the adjacent deformation region in order to establish the near-real kinematic allowable velocity field using the flow function method. Sezek et al. [19] set up the three-dimensional deformation of the plate rolling process utilizing the upper bound theorem. The kinematic allowable velocity field including the diffusion was derived by using the dual flow function. Hwang et al. [20] set up a theoretical model of asymmetric clad plate rolling using the flow function method and systematically discussed the effects of the curvature such as rolling speed ratio, roll diameter ratio, initial thickness ratio, plate flow stress ratio, total reduction, thickness ratio and rolling power. Jiang et al. [21] employed the flow function method and the upper bound method for bonding of cold-rolled clad plates to predict the post-rolling curvature, thickness ratio and rolling power. Liu et al. [22] innovatively put forward a three-dimensional kinematic tolerance velocity and strain rate field using dual flow function method and boundary conditions, and compared the calculated plate shapes with other models to prove the accuracy of the field. Zhang et al. [23] proposed a new exponential velocity and strain rate field to satisfy the kinematic tolerance conditions adopting the upper bound theorem. Che et al. [24] constructed the analytical model of the force and energy parameters of the gradient temperature rolling process by fusing the temperature field with the energy functional.

To sum up, a plate curvature model for the asymmetric rolling with different roll diameters is creatively built, and a new deformation boundary condition is put forward with the flow function method. The model was proved by simulation and experiments. The plate bending mechanism was obtained by the analysis of the relationship between the rolling power and plate curvature in this paper.

2 Theoretical parameter modeling

2.1 Geometric model

The following assumptions were made in order to model the plate curvature after rolling according to the actual rolling conditions:

The width spreading of the rolled clad plate after rolling is very small, so the three-dimensional model is deemed to two-dimensional plane strain model. The clad plate and work roll are deemed too rigid material. The friction coefficient is constant between the plate and the work roll, but the friction coefficient at the top roll m_1 is different from the friction coefficient at the bottom roll m_2. The flow direction of the plate at the inlet and the outlet of the roll seam is horizontal.

The geometric model of the double metal by asymmetric roll-bonding with the different diameters is shown in Fig. 1. The origin of the Cartesian coordinate system is located at the center of the bottom work roll. The x-axis points to the right, and the y-axis points to the top.

2.2 Two-dimensional flow function model

The deformation region is considered layer a and b, as indicated in Fig. 1, where a is the cladding layer and b is the base layer. The basis flow functions can be expressed as

$$\begin{cases} \phi_a^0 = \int_{y_j(x)}^{y} v_{xa}\, dy \\ \phi_b^0 = \int_{y_2(x)}^{y} v_{xb}\, dy \end{cases} \tag{1}$$

The volume of any section is the same according to the volume invariant criterion as follows:

$$\begin{cases} Q_a = v_{ia} t_{ia} = v_{xa} t_{xa} \\ Q_b = v_{ib} t_{ib} = v_{xb} t_{xb} \end{cases} \tag{2}$$

In Fig. 1, t_{xa}, t_{xb} can be represented by the boundary functions y_1, y_j, y_2 according to the geometric relations as follows:

$$\begin{cases} t_{xa} = y_1 - y_j \\ t_{xb} = y_j - y_2 \end{cases} \tag{3}$$

The basic flow function can be obtained through substituting Eqs. (2) and (3) into Eq. (1) as follows:

$$\begin{cases} \phi_a^0 = Q_a \dfrac{y - y_j}{y_1 - y_j} \\ \phi_b^0 = Q_b \dfrac{y - y_2}{y_j - y_2} \end{cases} \tag{4}$$

In order to satisfy the formula (5), $\phi^* a = 0$, $\phi^* b = 0$, the added flow functions of layer a and b require the factorization $(y-y_1)(y-y_j)$ and $(y-y_j)(y-y_2)$, respectively. After a large number of numerical simulation and experiments, it is verified that the horizontal velocity of the metal is close to a nonlinear distribution [24]. In order to satisfy the nonlinear distribution law, the shape function needs to contain the factor ax^2. In addition, the outlet velocity of the two layers on the bond surface is equal in order to combine the base and cladding metal after rolling, and the shape function also requires factor b. Consequently, the introduction of added flow field of parabolic

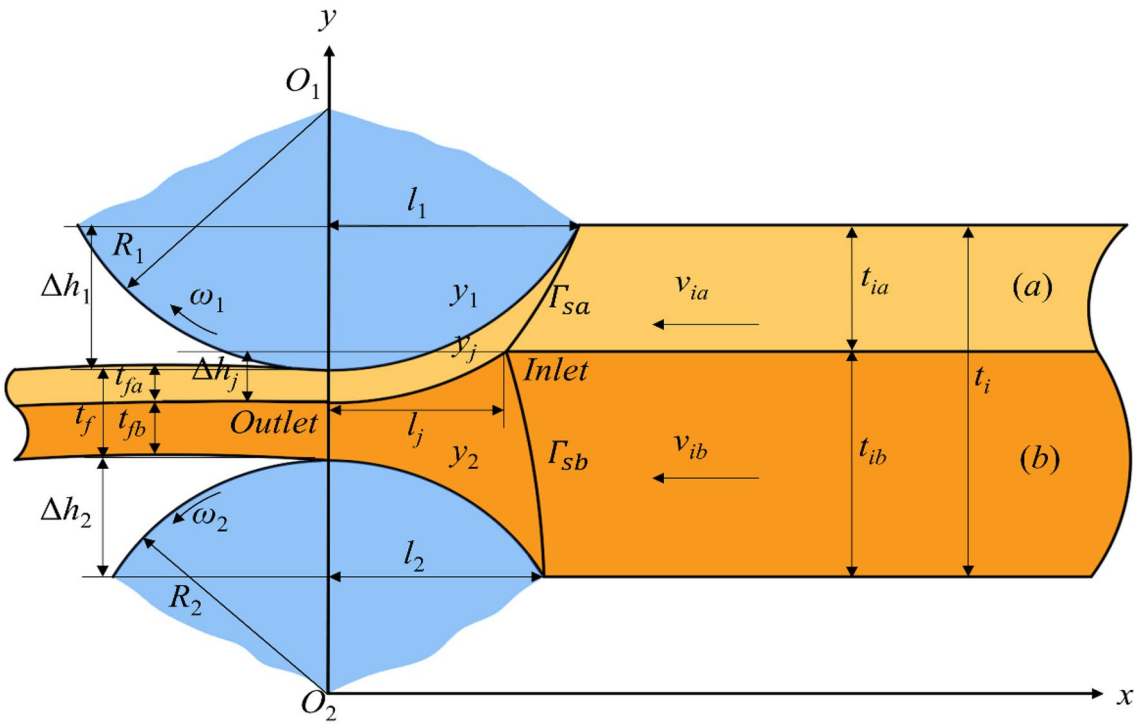


Fig. 1 Geometric model of asymmetric rolling process with different diameters

function $C_a=(a_1x^2+b_1)$ and $C_b=(a_2x^2+b_2)$ can avoid the error because of the assumption of homogeneous deformation and can converge to a more realistic motion velocity field.

The added flow functions of the layers *a* and *b* as follows:

$$\begin{cases} \phi_a^* = Q_a C_a (y-y_1)(y-y_j) \\ \phi_b^* = Q_b C_b (y-y_j)(y-y_2) \end{cases} \tag{5}$$

The complete flow function can be obtained by combining Eqs. (4) and (5) as

$$\begin{cases} \phi_a = Q_a[\dfrac{y-y_j}{y_1-y_j} + (y-y_1)(y-y_j)C_a] \\ \phi_b = Q_b[\dfrac{y-y_2}{y_j-y_2} + (y-y_j)(y-y_2)C_b] \end{cases} \tag{6}$$

2.3 Boundary function model

The contact interface and bonding interface can be expressed as follows:

$$\begin{cases} y_1 = R_2 + t_f + R_1 - \sqrt{R_1^2 - x^2} \\ y_j = R_2 + t_{fb} + R_j - \sqrt{R_j^2 - x^2} \\ y_2 = \sqrt{R_2^2 - x^2} \end{cases} \tag{7}$$

The bonding interface is similar to the circular function by numerical and experimental results. The bonding interface in the numerical element model, and the experiments are shown in Figs. 2 and 3.

The radius of the deformation region bond surface R_j is

$$R_j = R_1 + t_{fa} \tag{8}$$

The length of the deformation region is given by

$$\begin{cases} l_1 = \sqrt{2R_1\Delta h_1} \\ l_2 = \sqrt{2R_2\Delta h_2} \end{cases} \tag{9}$$

The geometry of the kinematic boundary conditions $y_3(x)$ and $y_4(x)$ is difficult to determine and cannot be obtained by assumption when determining Γ_{sa}, Γ_{sb}. In order to solve the expressions, the following equations are built depending on

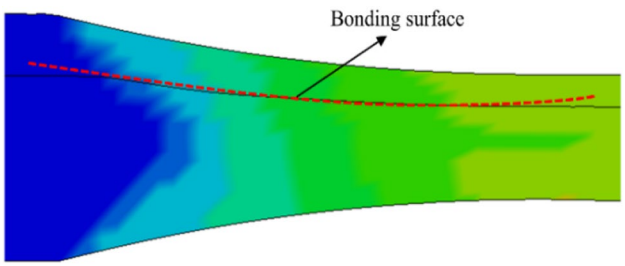


Fig. 2 The bonding interface of clad plate in the finite element model

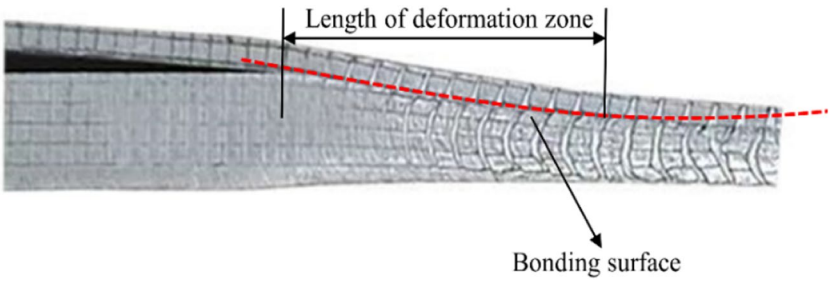


Fig. 3 The bonding interface of clad plate in the experiment

the real kinematic characteristics of the deformation region, incompressibility properties and the law of volume invariance which is given by

$$\phi_{ia} = \phi_a = \phi_{fa} \tag{10}$$

The specific expressions of ϕ_{ia}, ϕ_a and ϕ_{fa} are as follows:

$$\begin{cases} \phi_{ia} = \dfrac{y - y_j}{t_{ia}} \\ \phi_{fa} = \dfrac{y - y_j}{t_{fa}} \end{cases} \tag{11}$$

Combining Eqs. (10) and (11) yields

$$\frac{y - y_j}{y_1 - y_j} + (y - y_1)(y - y_j)C_a = \frac{y - y_j}{t_{fa}} \tag{12}$$

Considering that the speed interruption line is close to the inlet, there is

$$\lim_{x \to l}\left(R_1 - \sqrt{R_1^2 - x^2}\right) = \Delta h_1, l_1 = \sqrt{2R_1\Delta h_1} \tag{13}$$

In order to make the calculation process more convenient, the expressions for the velocity discontinuities Γ_{sa} and Γ_{sb} have been simplified to the y_3 and y_4 during the modeling process.

The expression of Γ_{sa} is obtained through combining Eqs. (12), (13), (7) and (8) as

$$y_3 = R_2 + t_f + \Delta h_1 + (1/t_{fa} - 1/t_{ia})/C_a \tag{14}$$

Similarly, the Γ_{sb} can also be derived by association as

$$y_4 = R_2 + t_{ib} - \Delta h_2 + (1/t_{fb} - 1/t_{ib})/C_b \tag{15}$$

2.4 Velocity and strain rate field

The partial derivative of the flow function can be obtained from the velocity field. Therefore, their respective horizontal velocity components can be obtained from the partial derivative of the ϕ_a and ϕ_b for y, and their respective vertical velocity components can be obtained from the partial derivative of the ϕ_a and ϕ_b for x. The velocity components v_{xa}, v_{ya}, v_{xb}, v_{yb} can be gained as follows:

$$v_{xa} = \frac{\partial \phi_a}{\partial y} = Q_a \cdot \left\{ \frac{1}{y_1(x) - y_j(x)} + 2C_a \cdot \left[y - \frac{y_1(x) + y_j(x)}{2} \right] \right\} \tag{16}$$

$$v_{xb} = \frac{\partial \phi_b}{\partial y} = Q_b \cdot \left\{ \frac{1}{y_j(x) - y_2(x)} + 2C_b \cdot \left[y - \frac{y_j(x) + y_2(x)}{2} \right] \right\} \tag{17}$$

$$v_{ya} = -\frac{\partial \phi_a}{\partial x} = -Q_a \cdot \left\{ \frac{\left[-y_j'(x)\right] \cdot \left[y_1(x) - y_j(x)\right] - \left[y - y_j(x)\right] \cdot \left[y_1'(x) - y_j'(x)\right]}{\left[y_1(x) - y_j(x)\right]^2} \right.$$
$$\left. + C_a' \cdot \left[y - y_1(x)\right] \cdot \left[y - y_j(x)\right] + C_a \cdot \left\{ \left[-y_1'(x)\right] \cdot \left[y - y_j(x)\right] + \left[y - y_1(x)\right] \cdot \left[-y_j'(x)\right] \right\} \right\} \tag{18}$$

$$v_{yb} = -\frac{\partial \phi_b}{\partial x} = -Q_b \cdot \left\{ \frac{\left[-y_2'(x)\right] \cdot \left[y_j(x) - y_2(x)\right] - \left[y - y_2(x)\right] \cdot \left[y_j'(x) - y_2'(x)\right]}{\left[y_j(x) - y_2(x)\right]^2} \right.$$
$$\left. + C_b' \cdot \left[y - y_j(x)\right] \cdot \left[y - y_2(x)\right] + C_b \cdot \left\{ \left[-y_j'(x)\right] \cdot \left[y - y_2(x)\right] + \left[y - y_j(x)\right] \cdot \left[-y_2'(x)\right] \right\} \right\} \tag{19}$$

The y_1', y_2' and y_j' represent the first partial derivative of y_1, y_2 and y_j' in Eqs. (18) and (19), respectively.

The following boundary conditions need to be satisfied which is given by

$$\begin{cases} v_{xa} = v_{xb} \big| x = 0, y = y_j \\ v_{ya} = v_{yb} \big| x = 0, y = y_j \end{cases} \tag{20}$$

Substituting Eqs. (16), (17), (18) and (19) into the deformation coordination condition Eq. (20), the relation between the parameters b_1 and b_2 for the layer a and b can be established as follows:

$$b_2 = \frac{t_{ia}\left(\frac{1}{t_f - t_{fb}} - b_1\left(t_f - t_{fb}\right)\right) - \frac{t_{ib}}{t_{fb}}}{t_{fb}t_{ib}} \tag{21}$$

The strain rate components is as follows:

$$\dot{\varepsilon}_{xa} = \frac{\partial v_{xa}}{\partial x} = Q_a \cdot \left\{ \frac{y'_j(x) - y'_1(x)}{[y_1(x) - y_j(x)]^2} + 2C_a' \cdot \left[y - \frac{y_1(x) + y_j(x)}{2} \right] - C_a \cdot [y'_1(x) + y'_j(x)] \right\} \tag{22}$$

$$\dot{\varepsilon}_{xb} = \frac{\partial v_{xb}}{\partial x} = Q_b \cdot \left\{ \frac{y'_2(x) - y'_j(x)}{[y_j(x) - y_2(x)]^2} + 2C_b' \cdot \left[y - \frac{y_j(x) + y_2(x)}{2} \right] - C_b \cdot [y'_j(x) + y'_2(x)] \right\} \tag{23}$$

$$\dot{\varepsilon}_{ya} = \frac{\partial v_{ya}}{\partial y} = -Q_a \cdot \left\{ \frac{y'_j(x) - y'_1(x)}{[y_1(x) - y_j(x)]^2} + 2C_a' \cdot \left[y - \frac{y_1(x) + y_j(x)}{2} \right] - C_a \cdot [y'_1(x) + y'_j(x)] \right\} \tag{24}$$

$$\dot{\varepsilon}_{yb} = \frac{\partial v_{yb}}{\partial y} = -Q_b \cdot \left\{ \frac{y'_2(x) - y'_j(x)}{[y_j(x) - y_2(x)]^2} + 2C_b' \cdot \left[y - \frac{y_j(x) + y_2(x)}{2} \right] - C_b \cdot [y'_j(x) + y'_2(x)] \right\} \tag{25}$$

$$\begin{aligned} \dot{\varepsilon}_{xya} &= \frac{1}{2}\left(\frac{\partial v_{xa}}{\partial y} + \frac{\partial v_{ya}}{\partial x} \right) \\ &= \frac{1}{2} Q_a \cdot \left\{ 2C_a + \left\{ \frac{-y''_j(x)}{y_1(x) - y_j(x)} - \frac{[y - y_j(x)] \cdot [y''_1(x) - y''_j(x)] - 2y'_1(x) \cdot [y'_1(x) - y'_j(x)]}{[y_1(x) - y_j(x)]^2} \right. \right. \\ &+ \left. \frac{2[y - y_j(x)] \cdot [y'_1(x) - y'_j(x)]^2}{[y_1(x) - y_j(x)]^3} - 2C_a' \cdot \right\} y'_1(x) \cdot [y - y_j(x)] + y'_j(x) \cdot [y - y_1(x)] \} \\ &- C_a \cdot \left\{ y''_1(x) \cdot [y - y_j(x)] + y''_j(x) \cdot [y - y_1(x)] - 2y'_1(x) \cdot y'_j(x) \right\} \\ &+ C_a'' \cdot [y - y_1(x)] \cdot [y - y_j(x)] \} \end{aligned} \tag{26}$$

$$\begin{aligned} \dot{\varepsilon}_{xyb} &= \frac{1}{2}\left(\frac{\partial v_{xb}}{\partial y} + \frac{\partial v_{yb}}{\partial x} \right) \\ &= \frac{1}{2} Q_b \cdot \left\{ 2C_b + \left\{ \frac{-y''_2(x)}{y_j(x) - y_2(x)} - \frac{[y - y_2(x)] \cdot [y''_j(x) - y''_2(x)] - 2y'_2(x) \cdot [y'_j(x) - y'_2(x)]}{[y_j(x) - y_2(x)]^2} \right. \right. \\ &+ \left. \frac{2[y - y_2(x)] \cdot [y'_j(x) - y'_2(x)]^2}{[y_j(x) - y_2(x)]^3} - 2C_b' \cdot \right\} y'_j(x) \cdot [y - y_2(x)] + y'_2(x) \cdot [y - y_j(x)] \} \\ &- C_b \cdot \left\{ y''_j(x) \cdot [y - y_2(x)] + y''_2(x) \cdot [y - y_j(x)] - 2y'_j(x) \cdot y'_2(x) \right\} \\ &+ C_b'' \cdot [y - y_j(x)] \cdot [y - y_2(x)] \} \end{aligned} \tag{27}$$

The y_1'', y_2'' and y_j'' represent the second partial derivative of y_1, y_2 and y_j'' in Eqs. (26) and (27), respectively.

The strain rate field needs to satisfy the volume incompressibility condition under the plane strain assumption. The strain rate field satisfies the volume incompressibility condition under the plane strain assumption according to Eqs. (22), (23), (24) and (25).

$$\begin{cases} \dot{\varepsilon}_{xa} + \dot{\varepsilon}_{ya} = 0 \\ \dot{\varepsilon}_{xb} + \dot{\varepsilon}_{yb} = 0 \end{cases} \tag{28}$$

2.5 Power model

The W_v, W_Γ and W_f can be gained by the five-node Gauss-Lejeune product formula. There are unknown parameters in the numerical expression of the total power a_1, a_2, b_1, Δh_1 and t_{fb}. The objective function is a function expression of total power, and the five unknown parameters a_1, a_2, b_1, Δh_1 and t_{fb} are taken as the independent variables of the optimization problem. The multivariate function nonlinear optimization problem is finally solved. The numerical solutions of the total power as follows:

$$W = W_{va} + W_{vb} + W_{\Gamma sa} + W_{\Gamma sb} + W_{fa} + W_{fb} \tag{29}$$

The plastic deformation power for the layers a and b is

$$\begin{aligned} W_{va} &= \frac{2\sigma_s}{\sqrt{3}} \left[\int_0^{l_j} dx \int_{y_j}^{y_1} (\dot{\varepsilon}_{\max} - \dot{\varepsilon}_{\min}) dy + \int_{l_j}^{l_1} dx \int_{y_3}^{y_1} (\dot{\varepsilon}_{\max} - \dot{\varepsilon}_{\min}) dy \right] \\ &= \frac{2\sigma_s}{\sqrt{3}} \left[\int_0^{l_j} v_y \Big|_{y_j}^{y_1} dx + \int_{l_j}^{l_1} v_y \Big|_{y_3}^{y_1} dx \right] \end{aligned} \tag{30}$$

$$\begin{aligned} W_{vb} &= \frac{2\sigma_s}{\sqrt{3}} \left[\int_0^{l_j} dx \int_{y_2}^{y_j} (\dot{\varepsilon}_{\max} - \dot{\varepsilon}_{\min}) dy + \int_{l_j}^{l_2} dx \int_{y_2}^{y_4} (\dot{\varepsilon}_{\max} - \dot{\varepsilon}_{\min}) dy \right] \\ &= \frac{2\sigma_s}{\sqrt{3}} \left[\int_0^{l_j} v_y \Big|_{y_2}^{y_j} dx + \int_{l_j}^{l_2} v_y \Big|_{y_2}^{y_4} dx \right] \end{aligned} \tag{31}$$

The shear power based on the velocity discontinuity is

$$W_{\Gamma sa} = k_a \int_{R_2 + t_{ib} - \Delta h_2}^{R_2 + t_i - \Delta h_2} \sqrt{(v_{xa} - v_0)^2 + v_{ya}^2} d\Gamma_{sa} \tag{32}$$

$$W_{\Gamma sb} = k_b \int_{R_2 - \Delta h_2}^{R_2 + t_{ib} - \Delta h_2} \sqrt{(v_{xb} - v_0)^2 + v_{yb}^2} d\Gamma_{sb} \tag{33}$$

The $d\Gamma_{sa}$ and $d\Gamma_{sb}$ are given by

$$\begin{cases} d\Gamma_{sa} = \sqrt{dx + (dy_3)^2} \\ d\Gamma_{sb} = \sqrt{dx + (dy_4)^2} \end{cases} \tag{34}$$

The $\vec{v}_0 \cdot \vec{n} = \vec{v} \cdot \vec{n}$ according to the normal velocity continuity condition, and the specific expression is shown as

$$\begin{cases} -v_0 \sin\alpha = -v_x \sin\alpha + v_y \cos\alpha \\ -v_0 \sin\beta = -v_x \sin\beta + v_y \cos\beta \end{cases} \tag{35}$$

According to the geometric relations in Fig. 1, it is known that

$$\begin{cases} \frac{v_{ya}}{v_{xa} - v_0} = tan\ \alpha = \frac{dy_3}{dx} \\ \frac{v_{yb}}{v_{xb} - v_0} = tan\ \beta = \frac{dy_4}{dx} \end{cases} \tag{36}$$

The final expression of the shear power model combining Eqs. (32), (33), (34) and (36) yields is

$$W_{\Gamma sa} = k_a \int_{R_2 + t_{ib} - \Delta h_2}^{R_2 + t_i - \Delta h_2} \frac{(v_{xa} - v_0)^2 + v_{ya}^2}{v_{ya}} dy \tag{37}$$

$$W_{\Gamma sb} = k_b \int_{R_2 - \Delta h_2}^{R_2 + t_{ib} - \Delta h_2} \frac{(v_{xb} - v_0)^2 + v_{yb}^2}{v_{yb}} dy \tag{38}$$

In this paper, the friction factor of the rolling process is calculated using Tarnowski's empirical formula, and the expression is given by

$$m = f\left[1 + \frac{1}{4}n(1-f)f^{\frac{1}{4}}\right] \tag{39}$$

The variable "n" indicates the ratio of the length of the deformation zone to the initial plate thickness, and the expression is $n = l/t_i$.

The frictional power generated by layers a and b with the work roll is as follows:

$$\begin{aligned} W_{fa} &= \int_{y_1} \tau_{f1} \left|\Delta v_f\right| dy_1 \\ &= m_1 k_a \int_0^{l_1} \sqrt{\left(v_{xa} - \omega_1 R_1 \sqrt{1-(x/R_1)^2}\right)^2 + \left(v_{ya} - \omega_1 R_1 \frac{x}{R_1}\right)^2} \sqrt{1+y'^2_1}\, dx \end{aligned} \tag{40}$$

$$\begin{aligned} W_{fb} &= \int_{y_2} \tau_{f2} \left|\Delta v_f\right| dy_2 \\ &= m_2 k_b \int_0^{l_2} \sqrt{\left(v_{xb} - \omega_2 R_2 \sqrt{1-(x/R_2)^2}\right)^2 + \left(v_{yb} - \omega_2 R_2 \frac{x}{R_2}\right)^2} \sqrt{1+y'^2_2}\, dx \end{aligned} \tag{41}$$

2.6 Curvature model

The curvature of the rolled plate is used in the study to quantify the bending of the rolled plate, and its value is the reciprocal of the bending radius of the plate. The shear and the axial strain can be obtained using the corresponding strain rate fields of Eqs. (22), (23), (26) and (27) as a transfer. Figure 4 represents the bending due to the difference in axial strains and shear strain difference.

The theoretical curvature model established in the text also can be applied to other double-layer metal clad plates when the work roll has the different diameters. The basic mechanical parameters of each layer of layered materials (such as the yield strength of the material and the friction coefficient between each layer of materials and the top and bottom work rolls, etc.) need to be replaced. The curvature can be calculated with the same method.

2.6.1 Curvature due to axial strain difference

The micro-elementary axial strain in the x-direction at any point within the deformation region for the layers a and b is

$$\varepsilon_{xa0} = \int_0^t \dot{\varepsilon}_{xa} dt \tag{42}$$

$$\varepsilon_{xb0} = \int_0^t \dot{\varepsilon}_{xb} dt \tag{43}$$

Along horizontal velocity sampling points are set every 1 mm in the rolling direction. The t can be obtained by dividing the length of the deformation zone by the average horizontal velocity at the same height.

The curvature of the rolled plate resulting from the axial strain difference according to the geometric relationship shown in Fig. 4 is

$$\frac{1}{r_a} = \frac{l_a - l_b}{t_f l_0} = \frac{\varepsilon_{xa} - \varepsilon_{xb}}{t_f} = \frac{\int_0^{l_1} \varepsilon_{xa0}\Big|_{y=y_1} dx - \int_0^{l_2} \varepsilon_{xb0}\Big|_{y=y_2} dx}{t_f} \tag{44}$$

2.6.2 Curvature due to shear strain difference

The micro-elementary shear strain at any point for the layers a and b can be expressed as

$$\gamma_{xya0} = 2\int_0^t \dot{\varepsilon}_{xya} dt \tag{45}$$

$$\gamma_{xyb_0} = 2\int_0^t \dot{\varepsilon}_{xyb} dt \tag{46}$$

The angular displacement of the clad plate can be calculated as follows:

$$\begin{aligned} \alpha_t = &\int_0^{l_j} dx \int_{y_j}^{y_1} \gamma_{xya0} dy + \int_{l_j}^{l_1} dx \int_{y_3}^{y_1} \gamma_{xya_0} dy \\ &+ \int_0^{l_j} dx \int_{y_2}^{y_j} \gamma_{xyb0} dy + \int_{l_j}^{l_2} dx \int_{y_2}^{y_4} \gamma_{xyb_0} dy \end{aligned} \tag{47}$$

The plate curvature due to shear strain difference is

$$\frac{1}{r_s} = \frac{\alpha_t}{l_1} = \frac{\int_0^{l_j} dx \int_{y_j}^{y_1} \gamma_{xya0} dy + \int_{l_j}^{l_1} dx \int_{y_3}^{y_1} \gamma_{xya_0} dy + \int_0^{l_j} dx \int_{y_2}^{y_j} \gamma_{xyb0} dy + \int_{l_j}^{l_2} dx \int_{y_2}^{y_4} \gamma_{xyb_0} dy}{l_1} \tag{48}$$

2.6.3 Total clad plate curvature

The total plate curvature is the sum of the curvature because of the shear strain difference and axial strain difference. The specific flow chart is shown in Fig. 5, and the final curvature can be expressed as

$$1/r = (r_s + r_a)/r_s r_a \tag{49}$$

3 Results and discussion

3.1 Validation of the flow function field

The streamline diagrams of the layers a and b are shown in Fig. 6. The metal materials are aluminum and magnesium respectively. The total thickness of the clad plate before rolling

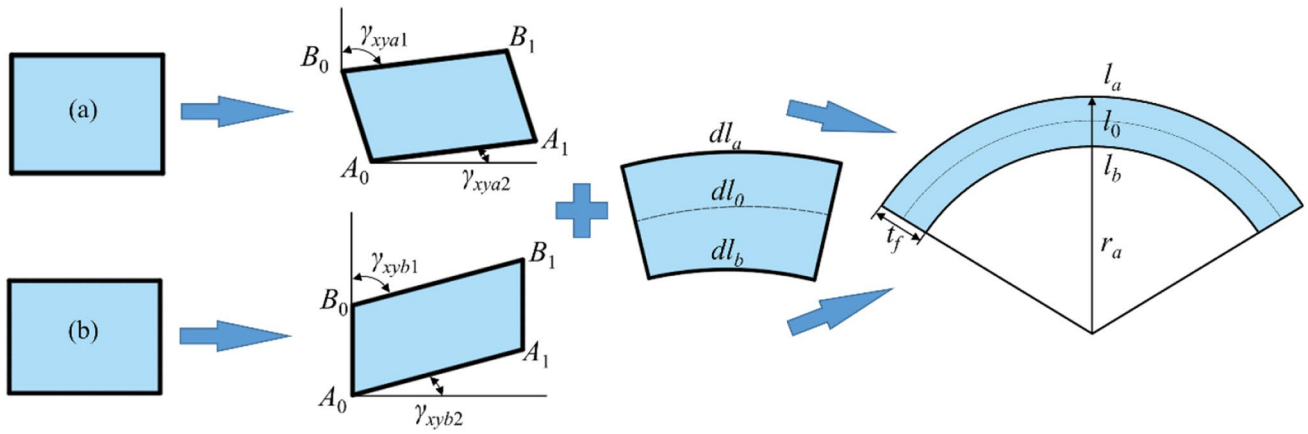
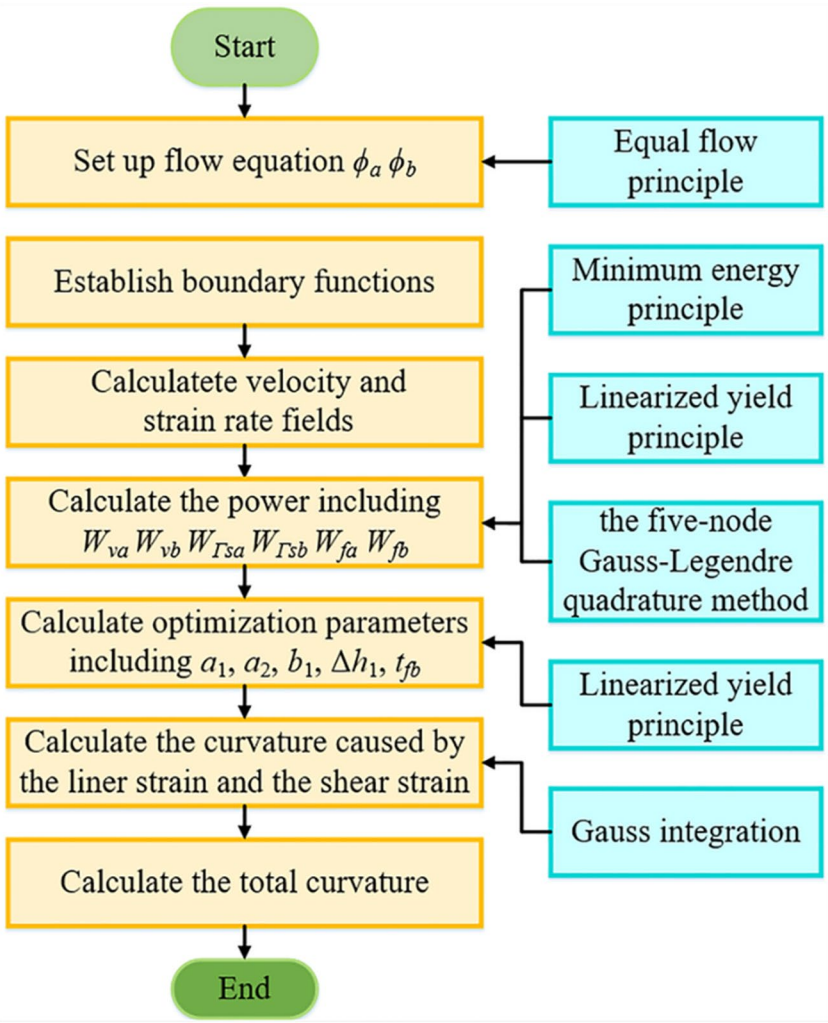


Fig. 4 Bending caused by axial strain difference and shear strain difference

Fig. 5 Solving flow chart



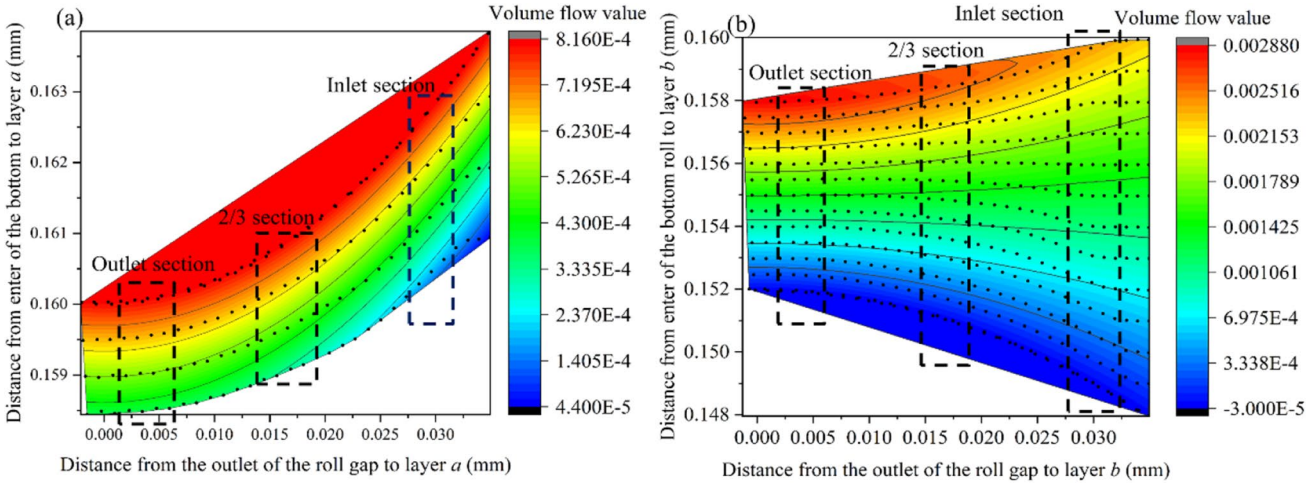


Fig. 6 **a** The streamline diagram of the layer *a* in the deformation region. **b** The streamline diagram of the layer *b*

was 16 mm, the radius of the top work roll was 160 mm, the roll diameter ratio was 1.05, the rolling velocity was 0.2 m/s, the initial thickness ratio was 0.25 and the rolling reduction was 8 mm.

In Fig. 6, the values of the streamlines shown represent the volume flow per unit area. The form of streamline conforms to the characteristics of metal flow by calculating the complete flow function field. The original volume flow ratio of layers *a* and *b* is 0.25. It can be seen from (a) and (b) in Fig. 6 that the average volume flow ratios of layers *a* and *b* at the inlet section, 2/3 section and outlet section are 0.28, 0.30 and 0.33, respectively. The average volume flow difference between layers *a* and *b* in the deformation region will cause the incongruity of metal deformation. The deformation at the inlet section began to appear uncoordinated, and the deformation is seriously uncoordinated at the outlet section. So the average volume flow difference of the outlet section has the greatest influence on the bending.

3.2 Validation of the velocity field model

A schematic diagram of the nodes selection method is demonstrated in Fig. 7. The group I, II, III and IV represent the velocity of the position on the inlet section, 1/2 section, 3/4 section and outlet section in the deformation region. The node number represents the plate thickness 4/5, 3/5, 2/5, 1/5 and the surface of the layer *a*, respectively. The comparison between the theoretical and the numerical results of the v_a is shown in Fig. 8. In Fig. 8, the Max deviation between the theoretical results of the v_a and the finite element simulation results is 1.28%, and the average deviation is 0.53%. Therefore, the theoretical model of v_a has adequate prediction precision.

The comparison between the theoretical and the numerical results of the v_b is shown in Fig. 9. The group I, II, III and IV represent the velocity of the position on inlet section, 1/2 section, 3/4 section and outlet section. The node number represents the plate thickness 5/5, 4/5, 3/5, 2/5 and 1/5 of the layer *b*, respectively. In these cases, the Max deviation between the theoretical results of the v_b and the numerical results is 1.0%, and the average deviation is 0.63%. The numerical curves in the deformation region are consistent with theoretical curves. Therefore, the theoretical model of v_b has adequate prediction precision.

The comparisons between the numerical and the theoretical values of v_a are shown in Fig. 10. The theoretical results of this v_a are close to numerical results. The deviation of the

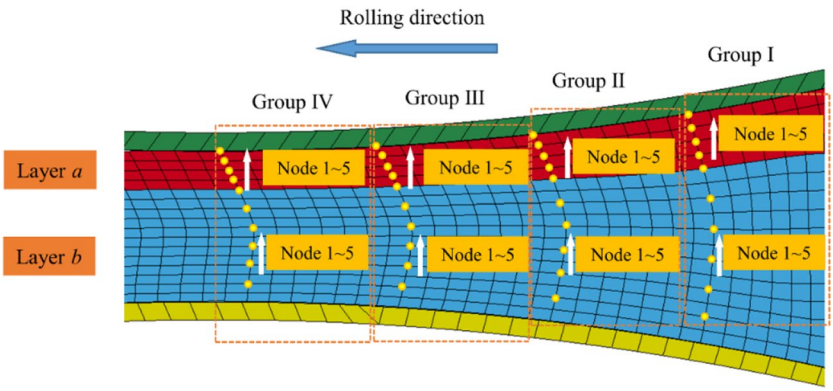


Fig. 7 Schematic diagram of the selected nodes for model verify

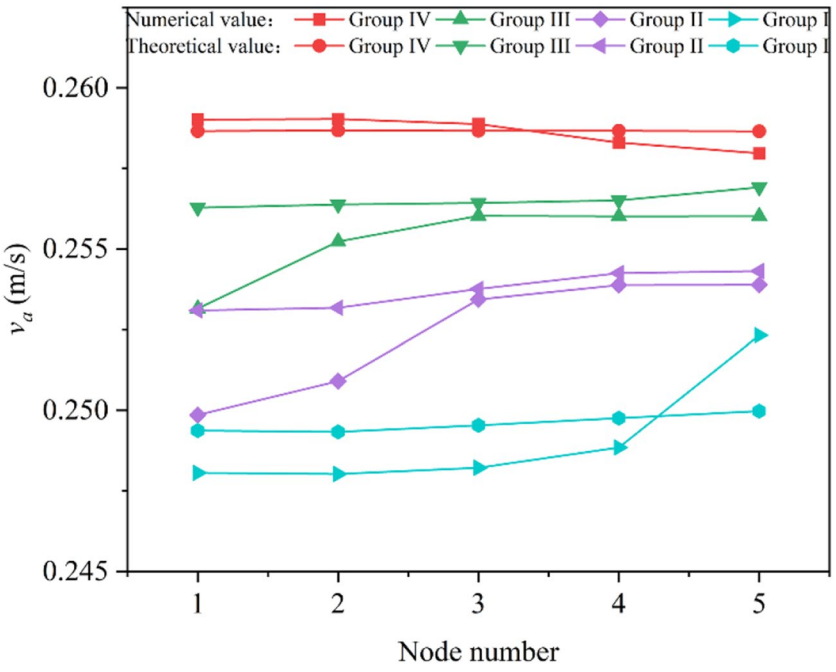


Fig. 8 The comparison between the theoretical and numerical results of the velocity of layer *a*

numerical results v_a is less than 5.0%, accounted for 92.1%, and that of less than 10%, accounted for 95.1%, compared with theoretical results. The velocity field model of layer *a* has adequate prediction precision.

The comparisons between the numerical and the theoretical values of v_a are shown in Fig. 11. The theoretical results of this v_b are close to numerical results. The deviation of the numerical results v_b is less than 5.0%, accounted for 81.1%, and that of less than 10%, accounted for 95.4%, compared with theoretical model calculation values. The velocity field model of layer *b* has adequate prediction precision.

To sum up, the singular points and other common numerical deviation would occur in the finite element calculation, which indicates the usage of the present theoretical velocity field to bring in the deviation. But overall, the velocity field model has adequate prediction precision.

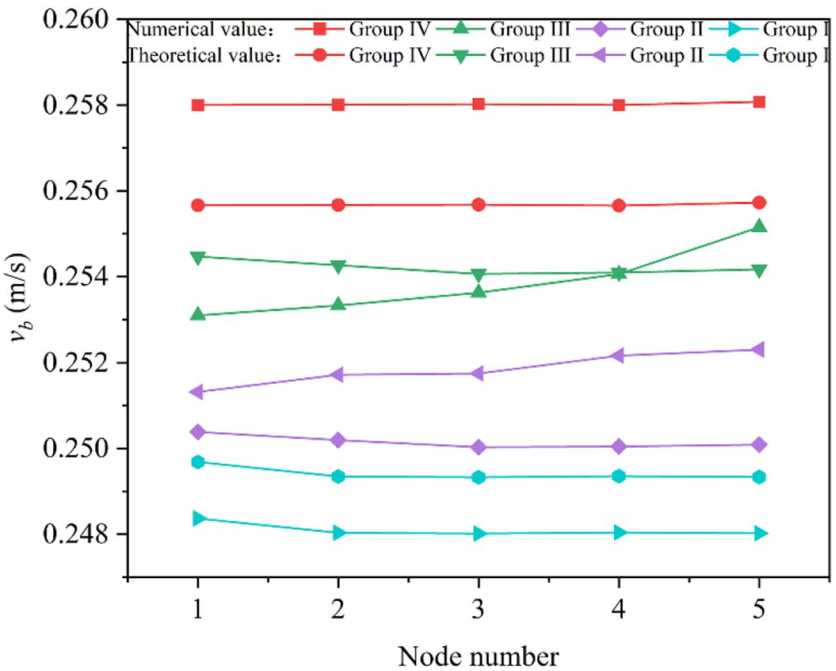


Fig. 9 The comparison between the theoretical and numerical results of the velocity of layer *b*

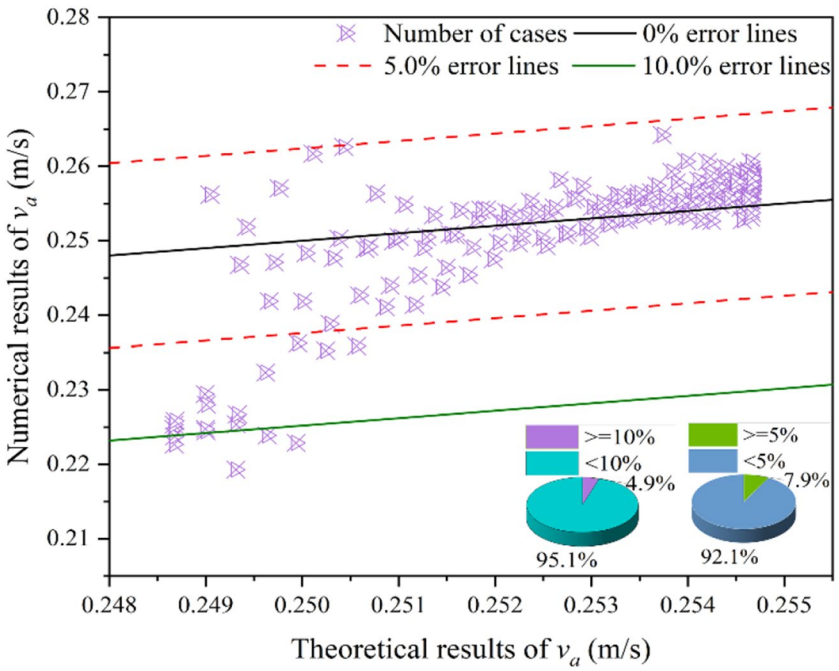


Fig. 10 Comparison of v_a between numerical results and theoretical results

3.3 Validation of the curvature model

In this paper, the ANSYS software is used to simulate the asymmetric rolling process with different process parameters. In the numerical process, the top and bottom rolls deformation are ignored and regarded as rigid bodies, and the diameter ratio is regarded the ratio of the top roll radius to the bottom roll radius.

The carbon steel/stainless steel (1200 °C) and magnesium/aluminum (400 °C) were used to confirm the precision of theoretical model of curvature. Combined with the above factors, the relevant rolling parameters are shown in Table 1, and the numerical simulation model is shown in Fig. 12.

3.3.1 Validation of model with different diameter ratios

The total plate thickness of magnesium/aluminum clad plate was 8 mm, with the original thickness ratio (the ratio of layer *a* thickness to total thickness) of 0.25, the reduction of

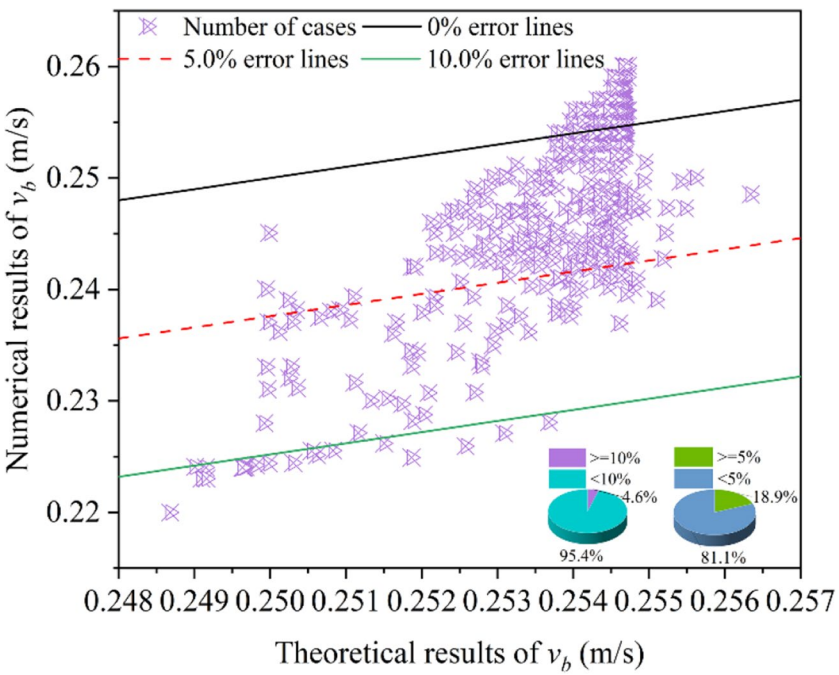


Fig. 11 Comparison of v_b between numerical results and theoretical results

Table 1 The rolling parameters

Item	Notations	Value
Yield strength of aluminum [Pa]	σ_{Al}	9.86×10^7
Yield strength of magnesium [Pa]	σ_{Mg}	9.62×10^7
Yield strength of stainless steel [Pa]	σ_{ss}	3.86×10^7
Yield strength of carbon steel [Pa]	σ_{cs}	2.29×10^7
Work roll speed [rad/s]	ω	1.6
The total thickness [mm]	t_i	8 ~ 32
Reduction [mm]	Δh	3.2 ~ 16.0
Top work roll radius [mm]	R_1	160
Work diameter ratios	i_r	1.00 ~ 1.20
Initial thickness ratios	i_t	0.25
Dynamic friction coefficient	μ	0.35
Static friction coefficient	f	0.4

4.0 mm, the radius of the top roll of 160 mm. Five different diameter ratios of 1.00, 1.01, 1.02, 1.05 and 1.20 were used to prove the accuracy of the model. The negative value of curvature indicates that it bends to the side of layer *b*. The numerical and theoretical results of the curvature under different diameter ratios are shown in Fig. 13.

In Fig. 13, the curvature that is because of the accumulated shear strain difference is negative values with the increase of the diameter ratio. The curvature calculated from the accumulated axial strain difference is positive. However, the influence of shear strain difference on the bending of rolled plates is greater than that caused by axial strain difference, so the total curvature can be changed from positive to negative, which will lead to the clad plate after rolling close to flat first and then reverse curved more and more. The deviation of the numerical peak curvature is 12.05% compared with theoretical curvature values, and the average deviation of the plate curvature is 7.75%. So the theoretical curvature model has adequate prediction precision.

The comparisons of shear stress on different positions of deformation with different diameter ratios are shown in Fig. 14a. The $\tau_{1/5}$, $\tau_{2/5}$, $\tau_{3/5}$, $\tau_{4/5}$ and $\tau_{5/5}$ represent the maximum shear stress of the position on inlet section, 2/5 section, 3/5 section, 4/5 section and outlet section. Through extensive simulation and theoretical analysis, the horizontal velocity of the plate in the rolling deformation region is the main factor affecting the overall velocity, and the vertical velocity is much smaller than the horizontal velocity, which can even be ignored [25].

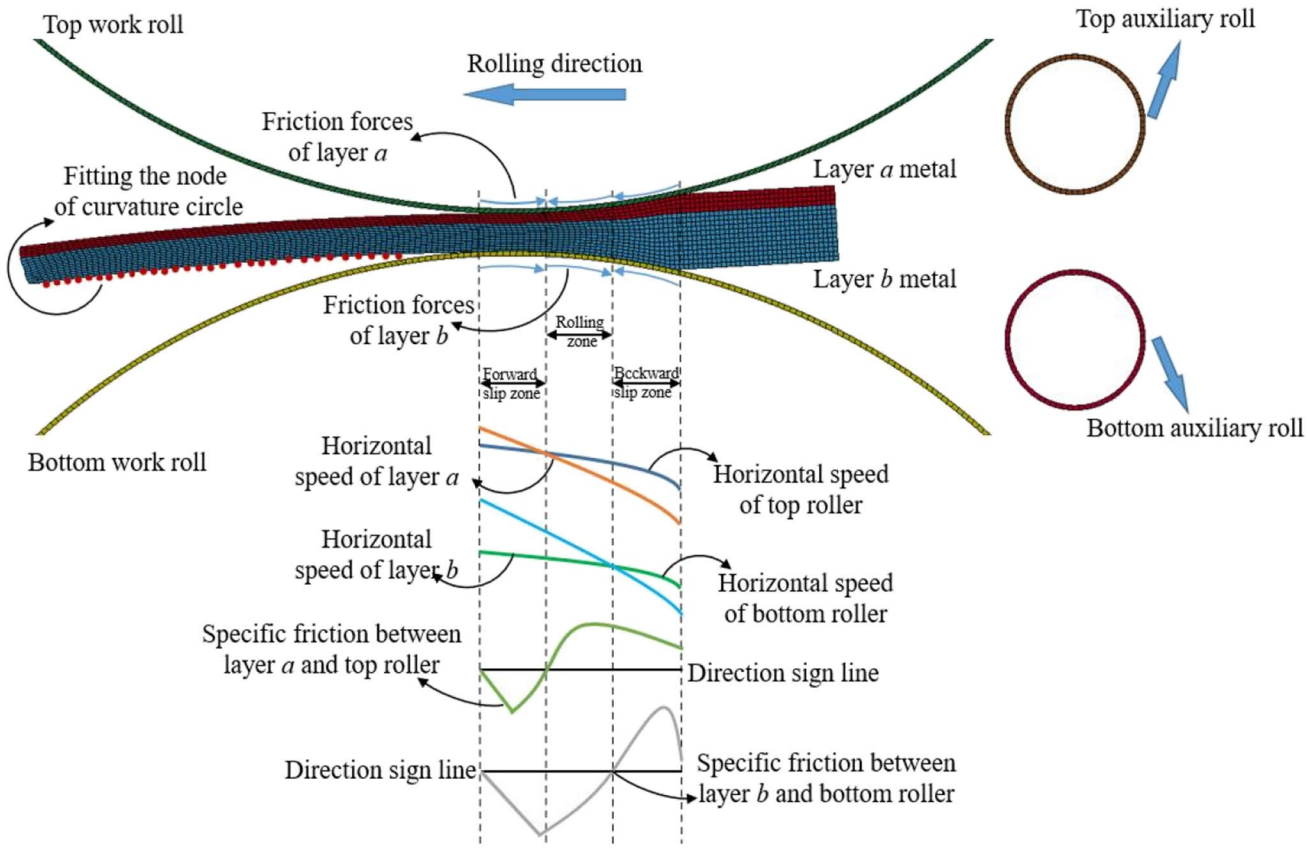


Fig. 12 The numerical simulation model

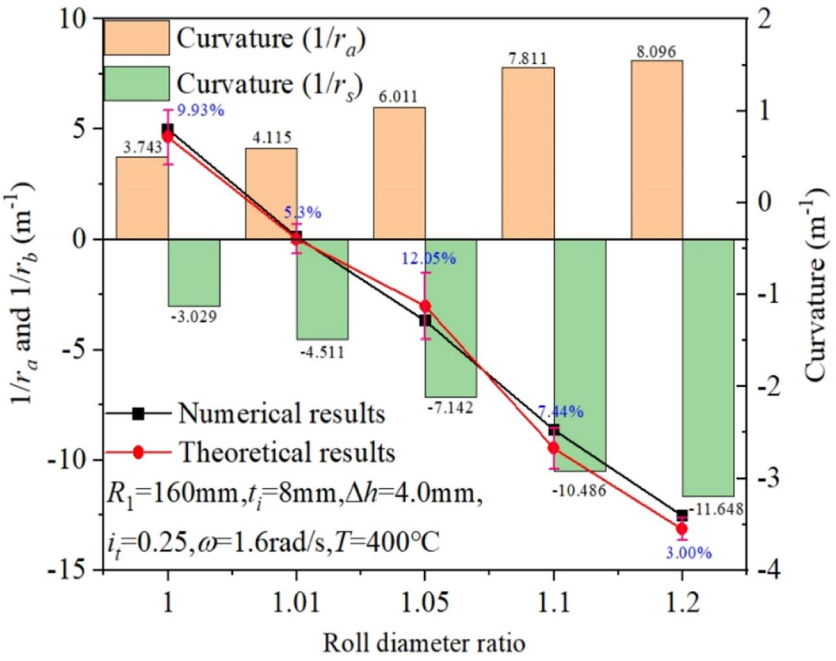


Fig. 13 Comparison of curvature results with different diameter ratios

In Fig. 14a, the shear stress increases with the increase of the diameter ratios. Analyzing the reasons according to the horizontal velocity field formulas (16 ~ 17), the decrease rate of velocity in layer *b* is faster than that in layer *a* with the increase of the diameter ratios. The velocity difference would increase, so the shear stress increases with the increase of the diameter ratios. The difference of shear stress between the layers *a* and *b* at the outlet section increases gradually, the shear stress of layer *b* is greater than that of layer *a*, and the yield strength of layer *b* is less than that of layer *a*, which causes greater bending deformation of the layer *b*. The deformation coordination ability of layers *a* and *b* is worse. So the plate shows bending phenomenon at the outlet section, and the velocity difference provides the shear stress required for bending.

The effect of rolling power on curvature with different diameter ratios is shown in Fig. 14b. The deformation and friction power contribute the most to the plastic deformation, and the power ratio of layer *b* is less than layer *a*. It is because the velocity difference decreases with the decrease of the diameter ratio. The shear power difference between

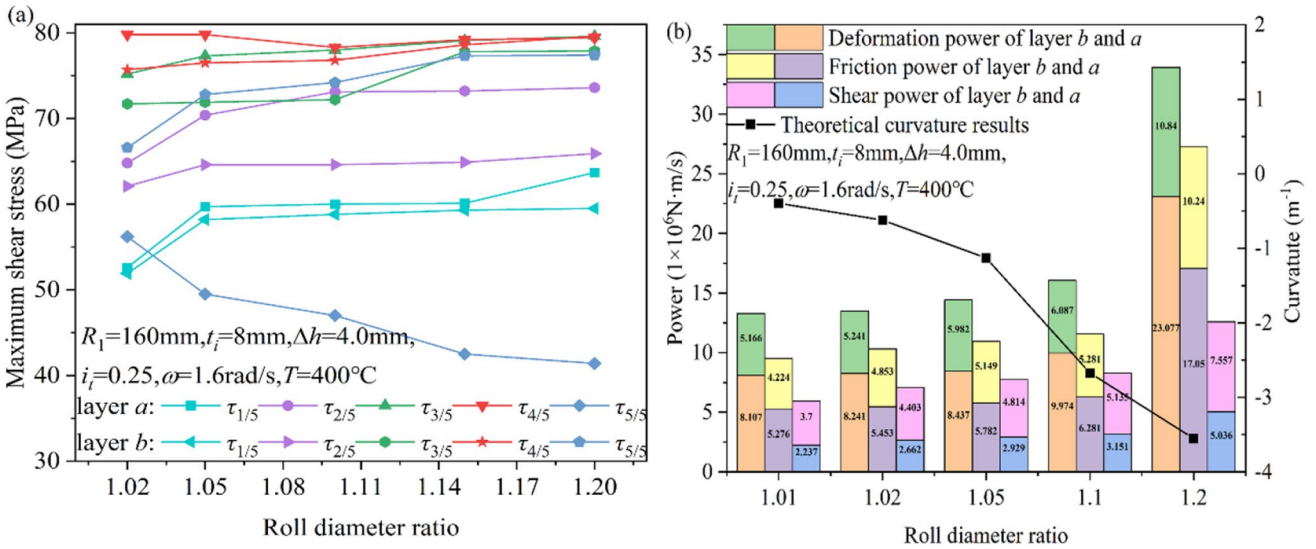


Fig. 14 **a** The shear stress with different diameter ratios. **b** The rolling power with different diameter ratios

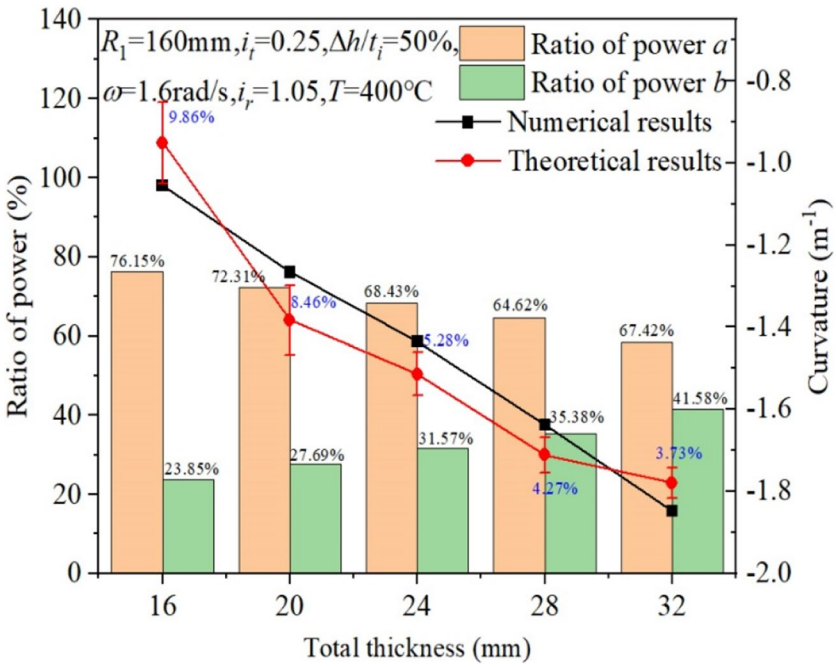


Fig. 15 Comparison of curvature results with different total thicknesses

layers *a* and *b* is the main reason for bending. In Fig. 14b, the shear power growth rate of layer *b* is faster than layer *a* with the increase of the diameter ratios. The shear power promotes greater bending deformation of the layer *b* with the increase of velocity difference. The clad plate coordinate deformation capacity will be worse and worse, resulting in the increasing degree of bending.

3.3.2 Validation of model with different total plate thickness

The bottom roll radius of magnesium/aluminum clad plate was 152 mm, with the top roll radius of 160 mm, the rolling temperature of 400 °C, the original layer thickness ratio of 0.25, the reduction rate of 50%. Five different total plate thicknesses

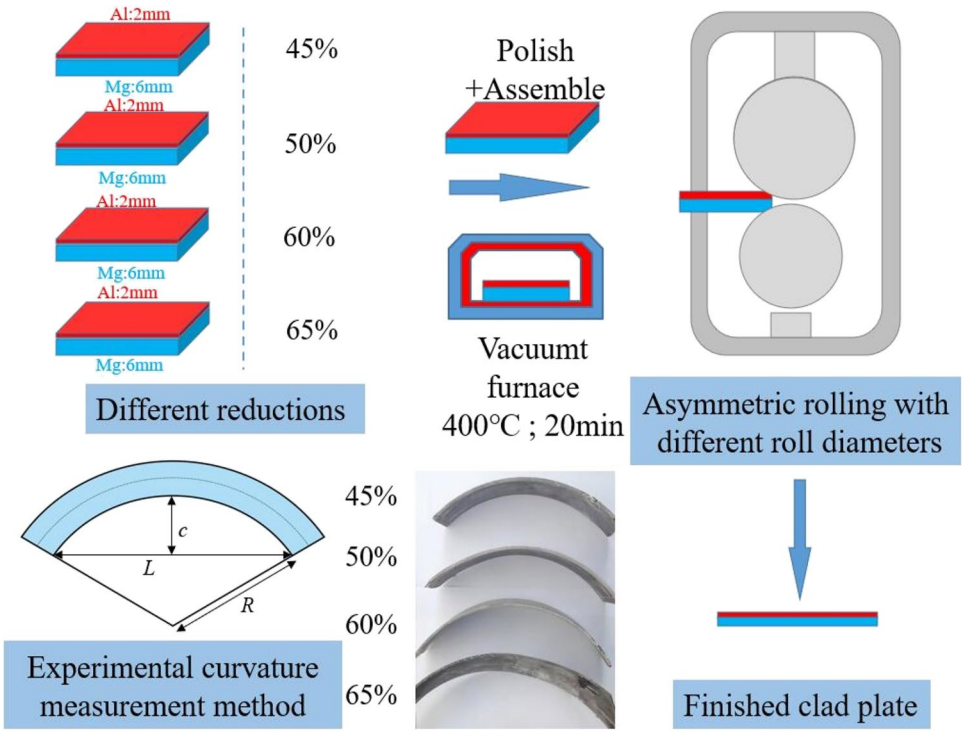


Fig. 16 The rolling bonding process of aluminum/magnesium with different roll diameters

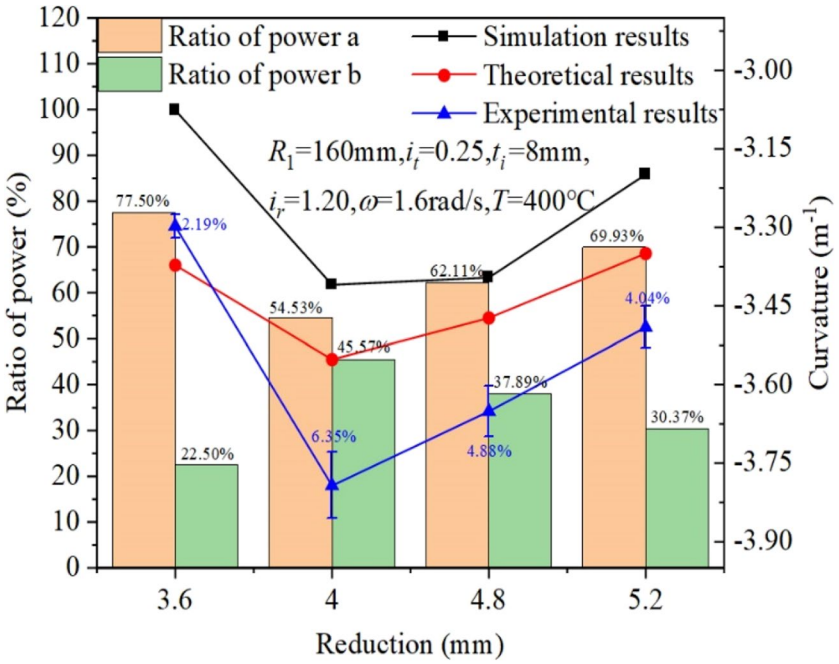


Fig. 17 Comparison of curvature results of aluminum/magnesium clad plate

of 16 mm, 18 mm, 24 mm, 28 mm and 32 mm were used to verify the accuracy of the model. The comparisons between the numerical and the theoretical results of the curvature for different initial plate thicknesses are shown in Fig. 15.

In Fig. 15, the absolute value of curvature decreases with the decrease of total thickness. Analyzing the reasons according to the theoretical model of velocity field formula, the thickness growth rate of layer *b* is faster than that of layer *a* with the increase of total thickness; the velocity difference would increase. So the velocity difference would change the ratio of power. It can be seen in Fig. 15 that the ratio of power of the layer *a* to the total power decreases from 76.15 to 67.42% with the increase of the total thickness before rolling, while the layer *b* increases from 23.85 to 41.58%. Based on the theoretical model of deformation region length, the extension length ratio of layers *a* and *b*

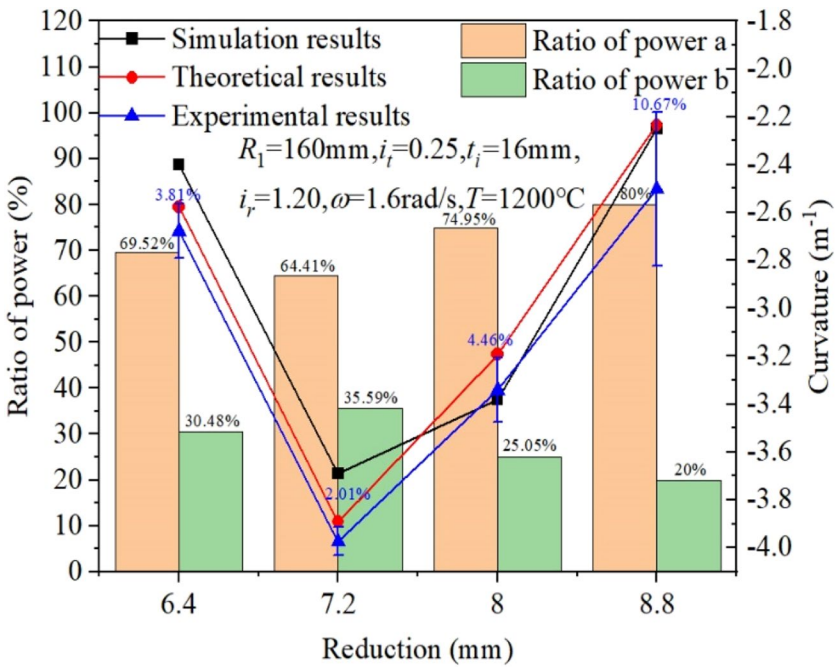


Fig. 18 Comparison of curvature results of carbon steel/stainless steel clad plate

decreases continuously with the increase of the total thickness. The strain speed of the layer *b* is faster compared to the strain speed of the layer *a*, making the clad plate coordinate deformation capacity worse, resulting in the increasing degree of bending of the clad plate. So the absolute value of curvature increases with the increase of the total thickness. The deviation of the numerical peak curvature values is 9.86% compared with theoretical curvature values, and the average deviation of the plate curvature is 6.32%, so the theoretical curvature model has adequate prediction precision.

3.4 Validation of model with experiment

The total thickness of aluminum/magnesium clad plate was 8 mm, with the original layer thickness ratio of 0.25, the top roll radius of 160 mm and the roll diameter ratio of 1.20. Four different reductions of 3.6 mm, 4.0 mm, 4.8 mm and 5.2 mm were selected to prove the precision. The rolling bonding process of aluminum/magnesium with different roll diameters is shown in Fig. 16. The bending radius after rolling is measured by graphic method, and $\rho = 1/R = 8c/(4c^2 + L^2)$ is used as the calculation formula of the curvature of the rolled plate, and the average value is taken after multiple measurements. The purpose of this is to effectively avoid the relative error in the experimental measurement process. The numerical, theoretical and experimental values of curvature with different reductions are shown in Fig. 17.

In Fig. 17, the absolute value of curvature first increases then decreases with the increase of the reductions. Analyzing the reasons according to the theoretical model of velocity field formula, the decrease rate of layer *b* is faster than layer *a* with the increase of the reductions; the velocity difference would increase. So the velocity difference would change the ratio of power. It can be seen in Fig. 17 that the ratio of power of the layer *a* to the total power first decreases from 77.50 to 54.53% then increases from 54.53 to 69.93% with the increase of the reductions, while the layer *b* has the opposite trend. Consequently, the ratio of power of the layer *b* to the total power first increases and then decreases; the strain speed of the layer *b* is first faster and then slower compared to the strain speed of the layer *a*. Based on the theoretical model of deformation region length, the extension length ratio of layers *a* and *b* first decreases continuously then increases with the increase of the reductions. The clad plate coordinate deformation capacity is first worse and then better; as a result, the absolute value of curvature increases first and then decreases. The Max deviation of the theoretical and experimental curvature results is 6.35%, and the average deviation is 4.37%. So the curvature model has adequate prediction precision.

The total thickness of the carbon steel/stainless steel clad plate was 16 mm, with the original layer thickness ratio of 0.25, the top roll radius of 160 mm and the roll diameter ratio of 1.20. Four different reductions of 6.4 mm, 7.2 mm, 8.0 mm and 8.8 mm were selected to verify the model accuracy. The numerical results, theoretical results and experimental results of curvature under different reductions are shown in Fig. 18. The Max deviation of the theoretical and experimental curvature results is 10.67%, and the average deviation of the plate curvature is 5.24%. Therefore, the curvature model has adequate prediction precision.

4 Conclusions

The article establishes the analytical models of the velocity field, strain rate field and shear strain rate, and the total powers are obtained by using upper bound method. The curvature model was ultimately established.

(1) The influence of rolling parameters is calculated by using the formulas. The results showed the shear power difference between layers *a* and *b* is the main reason for bending. The plate shows bending phenomenon at the outlet section, and the velocity difference provides the shear stress required for bending. The increase of the diameter ratios leads to the increase of the shear strain difference and shear power, and it leads to an increase in the total curvature of the rolled plate.

(2) The form of streamline conforms to the characteristics of metal flow by calculating the complete flow function field. The average volume flow difference between layers *a* and *b* in the deformation region would cause the incongruity of metal deformation. The average volume flow difference in the outlet section makes the deformation of layers *a* and *b* the most uncoordinated, so it has the greatest influence on the bending.

(3) The relative deviation of the numerical values v_a and v_b is less than 10%, accounting for 95.1% and 95.4%, compared with theoretical results. The accuracy of the clad plate curvature calculation model was proved by comparing theory, simulation and experiments, with minimum, maximum and average deviation of 2.01%, 12.05% and 5.92%. So the curvature model built is accurate.

(4) The change law of curvature under different conditions is studied. The bending degree of the clad plate increases when the diameter ratio or total plate thickness increases. However, it shows a trend of first increasing and then decreasing as the reduction increases. The process parameters are adjusted according to this variation pattern to control the subsequent required flatness rolling.

Author contribution Lian-Yun Jiang: conceptualization; writing—original draft preparation; visualization; project administration. Jia-Le Liang: writing—original draft preparation; visualization.

Ya-Fei Chen: formal analysis. Jian-Hui Shi: writing—review and editing; supervision.

Li-Feng Ma: visualization.

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Data availability Data and materials will be available upon request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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